MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10559325 (POWER SUPPLY, LOW VOLTAGE)

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Circuit Card Assembly: 10559325, (Power Supply, Low Voltage) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10559325	Circuit Card Assembly - Power Supply, Low Voltage

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PACKAGING DATA SHEET

10559325 Packaging of Circuit Card Assembly - (Power Supply, Low Voltage): 10559325

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The circuit card assembly shall be manufactured in accordance with Drawing F10559325 and drawings pertaining thereto.

3.2 General specification.- The circuit card assembly shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Part identification and marking

3.2.1 Electronic workmanship.- The circuit card assembly shall meet the workmanship requirements of Drawing F10559325 and drawings pertaining thereto.

3.3 Performance. - Unless otherwise specified, the circuit card assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (^{o}F) and 90 ^{o}F.

3.3.1 Loads and power.- The circuit card assembly shall perform as specified herein when the loads, auxiliary equipment, and power of Table I are applied as specified.

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TABLE I

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
1.	Loads:	Ohms (±5%)	Connected between the following pins of P1
1.1	Full load:		
1.1.1	Resistor	3.9	15A and 16B
1.1.2	Resistor	51	33A and 16B
1.1.3	Resistor	13	14B and 36B
1.1.4	Resistor	270	32B and 16B
1.2	Reduced load:		
1.2.1	Resistor	7.8	15A and 16B
1.2.2	Resistor	102	33A and 16B
1.2.3	Resistor	26	14B and 36B
1.2.4	Resistor	540	32B and 16B
1.3	Resistor	1800	10B and 5A
1.4	Capacitor (4 required)	.1 μF ±20% (ceramic)	32B and 16B 15A and 16B 33A and 16B 14B and 36B
2.	Power source:	17 to 31 Vdc adjustable	Applied between P1-4B (+) and P1-36B (-)
3.	Auxiliary equipment:	Preregulator switch drive circuitry as shown on Figure 1, or equivalent	Connected to P1 as shown on Figure 1

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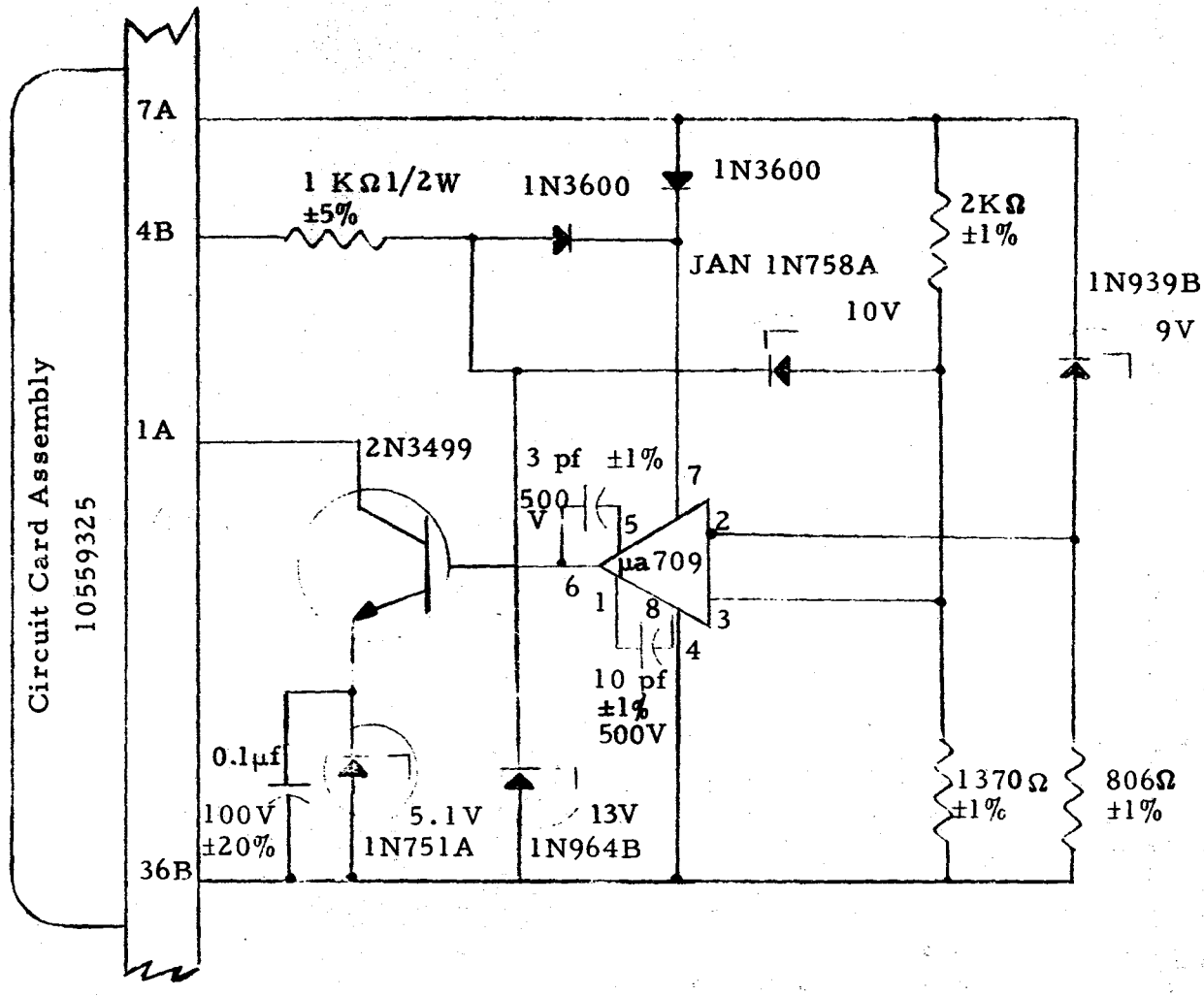


Figure 1. Preregulator Switch Drive

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3.3.1.1 Full-load output voltage and ripple. - With the input voltage at P1-4B set to 24 ±2 Vdc and with the full-load operating conditions specified in Table I (item 1.1), the output and ripple voltages at the output terminals shall be as specified in Table II.

TABLE II

OUTPUT (Pins of P1)	OUTPUT VOLTAGE (Vdc)	RIPPLE VOLTAGE (Volts p-p)
14B	15 ±1.2	0.9
15A	5.1 ±0.5	0.5
33A	-6 ±0.75	0.7
32B	29 ±2.5	2.0

3.3.1.2 Reduced-load output voltage. - With the input voltage at P1-4B set to 24 ±2 Vdc and the reduced-load operating conditions specified in Table I (item 1.2), the d-c voltage at the output terminals shall be as specified Table II.

3.3.1.3 Full-load input/output voltage regulation. - With the full-load operating conditions specified in Table I, the output voltage at P1-14B (+15 Volts) shall not vary more than ±0.5 Volt for an input voltage variation of 18 to 30 Vdc at P1-4B.

3.3.1.4 Sync signal. - With an input voltage of 24 ±2 Vdc at P1-4B and the full-load operating conditions specified in Table I, the output between P1-10B and P1-5A shall be a square wave having an amplitude of 9.2 ±1.5 Volts p-p at a frequency of 10 kHz ±30%.

3.3.2 Environmental.

3.3.2.1 Storage temperature. - The circuit card assembly shall meet the requirements of 3.3.1 at ambient temperature (60°F to 90°F) after exposure and thermal stabilization at -80°F and +160°F.

3.3.2.2 Operating temperatures. - The circuit card assembly shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which it shall meet the above requirements at ambient temperature (60°F to 90°F).

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3.3.2.3 Vibration.- The circuit card assembly shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

(a)	Amplitude:	0.5 inch maximum
(b)	Acceleration:	4g maximum
(c)	Crossover frequency:	12.5 Hz
(d)	Sweep time:	5 to 500 to 5 Hz in 15 minutes. Two sweeps in each of three mutually perpendicular axes

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three circuit card assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III, IV, and V herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table III, IV, and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests.- The requirements and tests in Table III shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

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TABLE III

(100% Inspection)

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Full-load output voltage and ripple	3.3.1.1	4.6.2
102.	Reduced load output voltage	3.3.1.2	4.6.3
103.	Full-load input/output voltage regulation	3.3.1.3	4.6.4
104.	Sync signal	3.3.1.4	4.6.5

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One circuit card assembly shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table IV.

TABLE IV

	CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings - Visual
302.	General specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental.- Three circuit card assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table V.

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TABLE V

CONTROL TEST		REQUIREMENT	TEST PROCEDURE
303.	Storage temperature	3.3.2.1	4.6.6.1
304.	Operating temperature	3.3.2.2	4.6.6.1
305.	Vibration	3.3.2.3	4.6.6.2

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions,

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and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment. - In conjunction with 4.5.2, the standard test equipment listed in Table VI shall be utilized in performing the applicable test of 4.6.

TABLE VI

	NOMENCLATURE	DESCRIPTION
1.	D-C Power Supply	17 to 31 Vdc, adjustable Ripple: 0.1 Volt rms maximum Source impedance: ≤0.01 ohm
2.	D-C Voltmeter	0-50 Volts, accuracy of ±1%
3.	Oscilloscope	Frequency response: d-c to 500 kHz Input impedance: 1 megohm or greater Sensitivity: 5 mV/cm Sweep time: 0.2 μs/cm
4.	Preregulator Switch Drive	As specified on Figure 1, or equivalent

4.6 Test methods and procedures.

4.6.1 Test conditions. - Unless otherwise specified, the test conditions shall be in accordance with the "Test Facilities" requirements of MIL-F-13926.

4.6.1.1 Initial conditions. - Connect the power source and auxiliary equipment to the circuit card assembly as specified in Table I. Unless otherwise specified, set the voltage of the power source to 24 ±2 Vdc.

4.6.2 Full-load output voltage and ripple. - Connect the loads specified in Table I for full-load operation. The output voltages and ripple at P1-14B, P1-15A, P1-33A, and P1-32B shall meet the requirements of 3.3.1.1..

4.6.3 Reduced-load output voltage. - Connect the loads specified in Table I for reduced-load operation. The output voltages and ripple at P1-14B, P1-15A, P1-33A, and P1-32B shall meet the requirements of 3.3.1.2.

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4.6.4 Full-load input/output voltage regulation. - Connect the loads specified in Table I for full-load operation. Vary the input voltage at P1-4B from 18 to 30 Vdc. The output voltage at P1-14B shall meet the requirements of 3.3.1.3.

4.6.5 Sync signal. - Connect the loads specified in Table I for full-load operation. Set the input voltage at P1-4B to 24 ±2 Vdc. The output waveform between P1-10B and P1-5A shall meet the requirements of 3.3.1.4.

4.6.6 Environmental tests. - The test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 except that the circuit card assembly may be mounted on the inside surface of the test chamber for the temperature test and the tolerance requirements for vibration frequency measuring instruments shall not apply.

4.6.6.1 Temperature test.-

NOTE: The rate of temperature change shall be less than 7°F per minute during temperature cycling.

(a) Place the circuit card assembly in the test chamber and reduce the chamber temperature to -80°F.
(b) Soak the circuit card assembly at this temperature for at least 5 hours with the unit not operating.
(c) Raise the chamber temperature to -40°F.
(d) Following thermal stabilization at -40°F, place the circuit card assembly in operation and perform the tests specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.5.
(e) Raise the chamber temperature to +160°F.
(f) Soak the circuit card assembly at +160°F for at least 5 hours with the assembly not operating.
(g) Reduce the chamber temperature to +125°F.
(h) Following thermal stabilization at +125°F, place the circuit card assembly in operation and perform the tests specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.5.
(i) Reduce the chamber temperature to ambient.
(j) Following stabilization at ambient temperature, place the circuit card assembly in operation and perform the tests specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.5.

The circuit card assembly shall meet the requirements of 3.3.2.1 and 3.3.2.2.

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4.6.6.2 Vibration. - With the circuit card assembly firmly mounted to the vibration table perform the following:

(a) Test in accordance with Method 514-I of MIL-STD-810 for category (f) equipment, except that the acceleration shall not exceed ±4g and the maximum excursion (double amplitude) shall be 0.5 inch.
(b) Sweep the frequency from 5 Hz to 500 Hz and back to 5 Hz in a period of 15 minutes. Perform two sweeps in each of the three mutually perpendicular axes.
(c) Following vibration cycling, examine the circuit card assembly for structural failures, loose hardware, wire breakage and abrasion.
(d) Place the circuit card assembly in operation and perform the tests specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.5.

The circuit card assembly shall meet the requirements of 3.3.2.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10559325. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The circuit card assembly is a functional module of the Laser Receiver-Transmitter, intended for use in the Laser Range Finder AN/VVS-1. Its functions are as follows:

(a) Provide regulated +5 and -5 Vdc output voltages.
(b) Provide a regulated +29 Vdc output voltage.
(c) Provide a regulated +15 Vdc output voltage.
(d) Provide a square wave sync signal output.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable packaging data sheet number (see 5.1).
(c) Selection of applicable levels of preservation, packaging, and packing.
(d) Applicable stock number.
(e) Requirement for First Article Submission.

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Project No. 1240-A146

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